The course is structured into 6 weeks, each focusing on a specific topic.

Week 1: Getting started with SQL

This week will let you learn the basics of SQL and databases. You will also learn how to query tables in a database.

Week 2: Introduction to relational Databases and Tables

This week is all about relational databases, creating tables, and modifying their contents.

Week 3: Intermediate SQL

In this module, you will learn more about different types of SQL queries, functions, string patterns, grouping, and sorting.

Week 4: Accessing databases with Python

This week, you will learn the nuances of accessing databases using Python libraries and SQL magic in Jupyter Notebooks.

Week 5: Course Assignment

This week is designed to give you an understanding of how to deal with real-world datasets and complete an assignment which tests your skills acquired throughout the course.

Week 6: Bonus Module: Advanced SQL for Data Engineers (Honors)

In this additional module, you will learn how to apply advanced queries in SQL, like Views, Stored Procedures, and ACID transactions.

Module 1: Getting Started with SQL

In this module, you will be introduced to databases. You will learn how to use basic SQL statements like SELECT, INSERT, UPDATE and DELETE. You will also get an understanding of how to refine your query results with the WHERE clause as well as using COUNT, LIMIT and DISTINCT.

Learning Objectives

- Define databases and SQL
- Write basic data retrieval queries using SELECT statement
- Refine query results by using COUNT, DISTINCT and LIMIT statements
- Insert data to a table using INSERT statements and alter or remove data from a table using UPDATE and DELETE statements

In this data-centric era, the role of data and the structures used to support its use become increasingly essential to humankind and the world at large. **Data science is a multi-billion dollar industry and** only poised to grow as we continue to identify its applications in the sciences, politics, business, and almost every other sector of our lives.

Analysts estimate the industry will grow to over 500 billion by the end of this decade.

As a data scientist, you can make $124,000 annually on average. Exciting, isn't it?

We generate and store trillions of gigabytes of data every day. With that being said, the need to analyze, manipulate, and, more importantly, manage this information has become a daunting task. And so, the importance of data science in today's world cannot be overstated.

As we move towards a data-centric world, the value of data science is rising to new heights. In the rapidly evolving digital landscape, databases are vast data repositories.

SQL, or Structured Query Language, **is the key that unlocks these treasure troves, enabling us to query, update, and manipulate data efficiently**. SQL plays a vital role in the data science workflow, enabling professionals to extract valuable insights from large, intricate datasets. SQL is powerful, globally relevant (in industry and academia), and widely used by most data scientists today.

A very versatile language called Python is used to **connect to various databases, execute SQL queries and retrieve results in a format conducive to further data analysis**. It offers a wide array of libraries that simplify the connecting process.

This course aims to **equip you with the knowledge and skills to harness the power of databases and SQL** in your data science endeavors effectively.

- In module 1 of this course, you will learn simple SQL for data manipulation in a database table.

- In module 2, you will learn about data definition statements and how to create our database tables.
- In module 3, you learn about built-in functions, sub-queries, and working with multiple tables.
- In module 4, you explore Python for accessing databases using Jupyter Notebooks.
- Module 5 primarily focuses on a project based on real-world data designed to test what you have learned from the course.
- In module, 6, covers advanced concepts like Stored Procedures, Views, ACID Transactions, Inner & Outer JOINs.

Additionally, integrated **hands-on labs and evaluations** allow you to apply the theoretical knowledge to real-world scenarios, enhancing your understanding and retention of the concepts. These are designed with the sole purpose of **maximizing your learning efficacy.** All this will make your journey from learning to mastering SQL for Data Science interactive, engaging, and, most importantly, effective.

Basic SQL

SELECT

COUNT DISTINCT LIMIT

INSERT

UPDATE and DELETE

Module 2: Introduction to Relational Databases and tables

In this module, you'll learn more about relational database concepts and their importance. This module helps you to understand the process of creating a table in your database on MySQL using the graphical interface and SQL scripts. Further, you will also learn how to alter the entries or delete the entries for any table in the database, or even delete the table itself.

Learning Objectives

- Describe relational database concepts such as entities, attributes, and primary keys
- Differentiate between DDL and DML statements
- Write commands to create tables in the database
- Explain how to alter, drop and truncate a table

Relational Database Concepts

Types of SQL Statements (DDL vs DML)

CREATE TABLE

ALTER, DROP, and Truncate Tables

SQL Scrips

Using IBM db2

Module 3: Intermediate SQL

This module helps you learn how to use string patterns and ranges to search data and how to sort and group data in result sets. You will also practice composing nested queries and execute select statements to access data from multiple tables.

Learning Objectives

- Use string patterns and ranges in SQL queries
- Group data into result sets
- Sort and order result sets
- Use in-built functions to refine query results
- Compose sub-queries and nested select statements

Refining your results

Using String Patterns and Ranges

Sorting result sets

Grouping results sets

Functions, Multiple Tables, and Sub-queries

Buil-in Database Functions

Date and tie Buil-in Functions

Subqueries and Nested Selects

Working with multiple tables

Module 4: Accessing Databases using Python

In this module you will learn the basic concepts of using Python to connect to databases. In a Jupyter Notebook, you will create tables, load data, query data using SQL magic and SQLite python library. You will also learn how to analyze data using Python.

Learning Objectives

- Access databases from Python using SQL magic
- Describe concepts related to accessing Databases using Python
- Create tables and insert data using Python
- Analyze a data set using SQL and Python

How to Access Databases Using Python

Writing code using DB-API

Accessing Databases wiht SQL Magic

Analyzing data wiht Python

Using IBM db2

Module 5: Course Assignment

n this module, you will be working with multiple real-world datasets for the city of Chicago. You will be asked questions that will help you understand the data just as you would in the real world. You will be assessed on the correctness of your SQL queries and results.

Learning Objectives

- Discuss real-world datasets
- Analyze table details and properties
- Interpret and translate data analysis questions into SQL queries
- Run SQL queries on real world datasets

Assignment Preparation: Working with real-world data sets and built-in SQL functions

Using IBM db2

Final Assignment

Final Exam and Course Wrap-up

Module 6: Bonus Module: Advanced SQL for Data Engineer (Honors)

This module covers some advanced SQL techniques that will be useful for Data Engineers. In this module, you will learn how to build more powerful queries with advanced SQL techniques like views, transactions, stored procedures, and joins. If you are following the Data Engineering track, you must complete this module. Completion of this module is not required for those completing the Data Science or Data Analyst tracks.

Learning Objectives

- Generate joins to query data from multiple tables
- Create and query a view
- Create stored procedures and invoke them from other code
- Describe the significance of ACID transactions
- Implement transactions in SQL statements
- Compare and contrast the benefits and disadvantages of using stored procedures
- Define views and describe the benefits they provide
- Compare and contrast the different JOIN operators

Views, Stored procedures, and Transactions

JOIN Statements

Quiz and Assignment for advanced SQL